

Optimising RAG Document Retrieval for Agronomic Advice

Section 1: Background and Context

Millions of smallholder farmers across Sub-Saharan Africa depend on timely, accurate agronomic advice to protect their harvests, yet the knowledge that could help them is often locked away in scattered extension manuals, research papers, and government guidance documents. When a pest outbreak, nutrient deficiency, or disease threatens a crop, the difference between the right information reaching a farmer quickly and it staying buried in an unsearched document can mean the difference between a saved harvest and a lost one. Retrieval Augmented Generation systems offer a way to close this gap, they work by first finding the most relevant documents for a question, then using an AI model to read them and compose an answer. But this only works if the retrieval step actually finds the right documents in the first place, if the correct information never surfaces, no amount of AI generation afterward can fix that. This project tackles that exact bottleneck, building and testing the retrieval engine that would sit at the core of such a system for agricultural extension advice.

Section 2: Core Problem

A farmer's question rarely uses the same language as an extension document. Someone might ask why their crop's leaves are turning yellow, while the relevant factsheet discusses nitrogen deficiency, the same issue described in completely different words. Within our knowledge base of 695 short agricultural factsheets, each question has several genuinely relevant documents, each with a different level of relevance. The real challenge is not just finding a document that looks related, but ranking the most correct ones highest. Making this harder, the knowledge base contains documents that look similar to the correct answer but are actually wrong, for example a document about potassium deficiency appearing for a question that is really about nitrogen deficiency. A system that only matches on keywords is easily fooled by these near miss documents.

Section 3: Current Approach and Limitations

Our starting point is TF-IDF retrieval, which matches documents to a query based on word overlap. This approach has clear limitations for this kind of data. It cannot recognise that two documents describing the same underlying issue in different words are related, it cannot tell the difference between a document that is topically similar but factually wrong and one that is genuinely correct, and it struggles with the varied and sometimes informal phrasing used in real farmer questions. Because our knowledge base deliberately contains these near miss documents, a purely keyword based approach is likely to rank several wrong documents highly, which is exactly the weakness this project is designed to address.

Section 4: Research Aim

The aim of this project is to build a retrieval system that ranks our 695 agricultural factsheets more accurately than a simple keyword matching approach, for a set of real farmer style questions. Performance is measured using nDCG at 5, based on how well our top 5 ranked documents match expert relevance judgements. We are aiming to meaningfully improve on baseline keyword matching, particularly on questions where the correct answer is conceptually related to the query but does not share obvious keywords, and on questions where near miss documents are present.

Section 5: Proposed Technical Approach

We plan to build and compare a few retrieval methods rather than trying everything available, developed and submitted through the Kaggle Notebook environment we are using. On the keyword matching side, we will use

BM25 as a refinement of basic TF-IDF matching. On the meaning based side, we will use dense embeddings from a pretrained sentence transformer model, such as all-MiniLM-L6-v2 or all-mpnet-base-v2, which represent the meaning of a question and a document as points in space, so that documents which are conceptually related but worded differently can still be matched correctly. We will also explore a hybrid approach that combines both methods, either through a weighted sum of BM25 and embedding similarity scores or through reciprocal rank fusion, since keyword matching and meaning based matching tend to catch different things. We are also considering whether an agriculture domain specific embedding model could improve results further, pending confirmation from the team on feasibility within our timeline. Our goal is to determine which of these approaches, or which combination, performs best against the graded relevance judgements, with particular attention to whether it can correctly avoid the near miss documents in our knowledge base.

Section 6: Scope and Boundaries

Our knowledge base is a fixed set of 695 factsheets covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation and fertiliser advice. We are working with this existing, self contained set of documents rather than sourcing or generating new ones. It is worth being clear about the boundary of this project, we are retrieving and ranking the correct documents, we are not generating the final written answer that would be given to a farmer.

Section 7: Evaluation Method

Performance is measured using nDCG at 5, based on how well our top 5 ranked documents per question match graded expert relevance judgements, ranging from 3 for a document that directly answers the question down to 0 for an irrelevant or near miss document. We will compare our BM25, embedding, and hybrid approaches against basic keyword matching using this metric.

Section 8: Significance

An improved retrieval system means that when a farmer or extension worker asks a question, the most accurate and relevant advice is far more likely to surface first, rather than a document that merely looks related. Given how directly agronomic decisions affect a household's harvest and food security, the difference between a correct and an almost correct document is not trivial, it can be the difference between fixing the right problem and wasting time and resources on the wrong one. A retrieval system that reliably avoids these near miss errors has real practical value as a foundation for future tools that could eventually deliver advice directly to smallholder farmers.